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An entropy weighted hybrid feature selection approach for heart disease prediction using machine learning

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Abstract

Heart disease still holds that top spot on the list of leading causes of death worldwide. Being able to predict heart disease at early stages really would help clinical practices work better and, in the end, it improves patient outcomes too. In this paper, we put forward an Entropy Weighted Hybrid Feature Selector (EWHFS), it is a kind of multi criteria feature selection framework, built with math backing, made for troublesome noisy and high dimensional biomedical datasets. The method blends Information Gain together with Fisher Score and also an Anti-Correlation Index, then it wraps everything into one entropy normalized optimization model. Overall, the model is set up to maximize feature relevance and class separability, but also reduce redundancy at the same time, not just one thing. We tested the approach on the UCI Heart Disease dataset, and we used stratified 10 fold cross validation, as usual. The experimental results show that EWHFS picked an optimal subset of 8 very discriminative features from 13 original clinical attributes. This cuts the dimensionality by 38.5% while still keeping the diagnostic information. So, the proposed approach reached about 94% accuracy, 93% precision, and 94% recall, and the F1 score came out as 93.5% while AUC was 0.95. It basically did better than the usual feature selection strategies, like Information Gain, Chi Square, Mutual Information, and even Recursive Feature Elimination. When we compare it with the strongest baseline, EWHFS bumped the classification accuracy up roughly 3–6%, yet it still kept pretty steady behavior across the validation folds. Then for the statistical validation, paired t tests plus Wilcoxon signed rank tests were used, and they verified that the gains were significant, with $p < 0.05$. These results demonstrate that EWHFS provides an effective, interpretable, and computationally efficient solution for heart disease prediction and clinical decision support systems.

Keywords Heart disease prediction, Entropy weighted hybrid feature selection (EWHFS), Information gain (IG), Fisher ratio class separability (FS), Anti correlation index (ACI), UCI dataset

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1 Introduction

Cardiovascular disease (CVD), which includes coronary heart disease (CHD), stroke, and heart failure, remains the world's leading cause of mortality, accounting for an estimated 19.8 million deaths globally in 2022, corresponding to approximately 32% of all global deaths, as reported by the World Health Organization [1]. The worldwide rate of premature death from heart disease is demonstrated by the fact that almost 85 percent of deaths result from myocardial infarction and stroke according to evidence from multiple studies [1, 2]. CVD affects all countries because it exists outside high-income nations which experience its most severe impact due to their insufficient healthcare programs that stop people from dying early through better diagnostic procedures and health services [2]. The present system for early detection requires a link between its operations and accurate risk assessment methods because both components function as vital resources for reduction of CVD health problems and associated death rates.

The Framingham Risk Score and QRISK cardiovascular risk assessment models show limited effectiveness in diverse population groups because these models depend on linear predictions and their designs fail to detect complex interactions that involve multiple clinical risk factors [3, 4]. In recent years Artificial Intelligence and Machine Learning methods have become effective tools for predicting cardiovascular risks because these techniques can analyse complex nonlinear patterns in extensive medical records to detect high risk patients earlier than traditional statistical methods [5, 6]. The AI driven models display superior predictive accuracy which enables them to forecast both in hospital cardiac arrests and future cardiovascular events because they can detect small details that exist within electronic health records. [7, 8].

Although the present machine learning cardiovascular prediction models show promise yet their performance faces critical challenges that limit their operational capabilities. The majority of studies use small and uniform datasets which causes their results to overfit and their findings to be unapplicable to different patient groups [9]. The usual feature selection approaches, like Mutual Information and Chi Square, plus other univariate filter-based techniques, tend to score the features separately, without accounting for duplicate features and the interactive influence that often appears in clinical datasets [10, 11]. Deep learning models provide high predictive performance but since they cannot really explain the outcome in a straightforward way, they are hard to rely on in medical settings where understandable results are needed [12]. Meanwhile the research that is already out there mostly tries to push classification accuracy, but it does not properly apply feature selection strategies that check multiple criteria, so they may end up picking redundant indicators, or so to speak unnecessary measurements, which have little clinical relevance and can also reduce model performance and transparency [13].Several studies suggest that mixing hybrid and ensemble learning approaches can raise the accuracy of cardiovascular disease predictions. Still, these gains tend to come with two major problems, mainly because they require more computing resources than usual, and because their results become harder to interpret, which then slows down adoption in healthcare environments that run with tight budget and staffing limits. [14, 15]. In this work we introduce an Entropy Weighted Hybrid Feature Selector (EWHFS), and it basically relies on an entropy driven adaptive weighting setup to look at the data. It does that by using information gain evaluation, class separability checking, plus a redundancy test step, all in one go. With its single scoring system, EWHFS helps pick a feature subset

that is both compact and information-rich. At the same time, it aims to keep strong discriminative power between the classes, so coronary artery disease prediction can be handled more accurately. Overall, the approach supports better classification outcomes, while also improving result understanding and expanding the system's practical usefulness, which is meant to support trustworthy decision tools for medical professionals.

The primary contributions of this work are summarized as follows:

- Development of a fresh Entropy measure to sort out, those cardiovascular risk factors that are unusually informative.
- Entropy Normalized Multi Criteria Feature Fusion: Unlike conventional entropy-based approaches that mix feature relevance measures in a kind of heuristic way, EWHFS introduces an entropy normalized scoring fusion mechanism that helps ensure scale invariant, and pretty stable feature ranking across heterogeneous clinical attributes, especially when you're juggling different data scales.
- Explicit Redundancy Suppression through Anti Correlation Index (ACI): The proposed framework uses a math defined redundancy penalty, based on an Anti-Correlation Index (ACI) type idea. It basically helps suppress explicit redundancy by minimizing feature dependency, and it tries to block the selection of those highly correlated attributes, sort of prevent overlap, so you end up with a smaller feature subset that stays compact, and non-redundant.
- Improved Predictive Performance and Interpretability:
- By simultaneously optimizing feature relevance class separability and redundancy minimization, EWHFS boosts classification performance while still keeping clinical interpretability, which is why it can be used in healthcare decision support systems.
- Comprehensive Experimental Validation:
- The effectiveness of EWHFS gets validated on the UCI Heart Disease dataset via stratified 10-fold cross validation, and then a comparative look against more conventional feature selection methods. In the end, the proposed framework turns out better, with higher accuracy, precision, recall, the F1-score, and AUC metrics, which sort of confirms both robustness and generalization ability even when you vary the splits.

2 Related work

Different techniques of feature selection have been proposed by researchers to predict heart diseases.

1. Sharma and his team created a hybrid feature selection system through their entropy-based approach which combined Information Gain with Mutual Information and Entropy Rank Weighting to identify the most effective cardiovascular predictors from the UCI heart dataset. The method achieved two outcomes because it decreased the number of dimensions and improved model transparency which led to a 4.3 percent accuracy increase when used with Random Forest and XG-Boost classifiers beyond traditional filter-based techniques [16].
2. Patel and Kaur developed a hybrid entropy correlation model which used Pearson correlation coefficients together with entropy scoring to eliminate unnecessary and irrelevant attributes. This approach delivered better classification accuracy through

Gradient Boosting and Support Vector Machine models when compared to Chi-square and Relief-F methods [17].

3. Reddy et al. came up with an Entropy Assisted Genetic Wrapper Selector, which is a combination of using entropy for ranking features and genetic search for optimizing selection of attributes. This model got excellent improvements in both precision and recall, especially for the multi-class cardiovascular disease scenario. Moreover, it beat the well-known selectors like Recursive Feature Elimination and PCA-based reduction in its performance [18].
4. Chandra and his colleagues came up with an ensemble-based entropy feature weighting scheme that allocated entropy scores to different feature subsets in order not to overfit. The authors proved that the use of entropy-based feature selection made classifiers more confident with the logistic regression and neural networks used for heart disease prediction [19].
5. Zhang and his colleagues' research featured a method that utilized the combination of SHAP and entropy weights to boost the model's ability to be explained as well as its clinical interpretability by making the most influential cardiovascular risk factors such as age, cholesterol and resting blood pressure, etc. Their hybrid entropy SHAP method achieved a test accuracy of 97% and showed better stability in the presence of dataset imbalance issues [20].
6. Johnson and his colleagues have come up with a multi stage entropy scoring algorithm that merged Relief-F scoring with mutual information filtering for cardiovascular feature enhancement. The approach of researchers removed noisy and redundant predictors like fasting blood sugar and resting ECG, which led to a 6% accuracy improvement when using an SVM-RF hybrid classifier over single stage selection methods [21].
7. In the paper published by Liu et al., a Deep Neural Network model improved by entropy -based dimensionality reduction was assessed for its ability to diagnose coronary artery disease with the Z-Alizadeh dataset as the primary source. Their entropy DNN pipeline could attain specificity as high as 95%, thereby surpassing the performance of PCA-based deep learning approaches, and thus illustrating the superiority of entropy over variance in this context [22].
8. Mishra and Sharma introduced a new feature scaling and normalization method based on entropy ranking, together with XGBoost, to enhance the heart disease prediction process. As a result of their strategies, the training time was decreased by 24% while the AUC values were raised to 0.971 in the Cleveland dataset from this result represents a major improvement over the Chi2 and ANOVA feature selection methods [23].
9. A mixed entropy correlation weighted feature selection model was suggested by Mathew et al. that was able to change feature relevance scores based on the interdependence of medical attributes like cholesterol and blood pressure. The methodology has emerged as the second-best technique for enhancement of sensitivity, which reached 8% when tested on SVM and GBM classifiers. [24].
10. The researchers used entropy weighted SHAP value ranking to find out which factors most affected cardiovascular health. The researchers identified four key factors which included age, chest pain type, ST slope, and exercise induced angina. The researchers used entropy weighted SHAP value ranking to determine which

factors had the greatest impact on cardiovascular health. Their integrated method not only exemplified a 97.5% accuracy in prediction but also generated output which was understandable in clinical context thereby supporting the decision making of cardiologists [25].

11. Al Qahtani and his colleagues proposed a model combining deep CNN with an entropy-weighting strategy. This model was trained on a massive clinical dataset from hospitals for the classification of heart abnormalities into multiple classes. Their research not only demonstrated a great ability of the model to generalize but also used entropy stability conditions to reduce overfitting, resulting in 5% accuracy gain over standard CNNs [26].
12. Jaiswal and Patel's treatment of instability and redundancy in classical algorithms through the combination of genetic optimization with an entropy enhanced wrapper selector was very successful indeed. Their model was able to boost XGBoost accuracy by 8.6% and displayed extremely robust performance over ten folds of cross validation outdoing RFE and Filter Gain methods [27].
13. Olatunji and others have suggested a model for adaptive entropy learning that will be combined with Light GBM in order to classify the levels of heart disease severity. Their technique managed to accomplish a precision of 0.98 and also proved that adjusting weights with the help of entropy decreased the bias in imbalanced datasets to a great extent when compared to SMOTE based oversampling [28].
14. The researchers known as Banerjee and others came up with a multi entropy fusion model which made use of Shannon, Tsallis, and Rényi entropy statistics combined together to enhance the feature ranking stability. Their method was able to score 7% F1-score higher than the classical entropy methods and also yielded more consistent classification across the two datasets namely Kaggle and Z-Alizadeh [29].
15. Chouhan et al. have come up with an innovative hybrid method that combines entropy and bagging, where the attributes ranked by entropy were used to create several tree-based classifiers that form the base of the bagging ensemble. Their experimentation demonstrated better ROC and PR curve metrics than AdaBoost and the independent Random Forest models, thereby validating the entropy-based diversity enhancement [30].
16. Al Mahdi et al. proposed a heart disease prediction framework that combines Genetic Algorithm (GA) based feature selection with an optimized ensemble deep learning model. Their approach employed Tunicate Swarm Optimization (TSA) to determine optimal ensemble weights and achieved 97.5% accuracy, demonstrating the effectiveness of integrating feature selection with ensemble learning for cardiovascular diagnosis [32].
17. Kumar et al. presented this kind of comprehensive review about machine learning methods for heart disease prediction and they seem to also stress the growing importance of hybrid feature engineering, explainable AI, federated learning, and deep learning frameworks in healthcare use cases. In a sense their work points toward why interpretable and more generalizable feature selection strategies are needed for clinical deployment, and it's kind of important to not lose that meaning when you actually apply it [33].
18. Thatha et al. introduced a Red Fox Optimization –based feature selection framework for cardiovascular risk prediction, using Cleveland and HARP datasets. Their model,

showed better predictive stability, and also reduced overfitting a bit. At the same time, it achieved superior overall results in terms of accuracy, recall, F score, and AUC [34].

19. Kailasanathan et al. developed an interpretable machine learning framework, for heart disease prediction using feature sensitive learning and explainable decision mechanisms kind of. Their results showed that when you combine predictive accuracy with transparency it really helps, the practical applicability of cardiovascular diagnostic systems a lot [35].
20. Tan et al. proposed this optimized feature selection, and deep neural network kind of framework for cardiac disease prediction. In their work they kind of showed that joining feature optimization together with deep learning architectures improves classification results and it also makes the whole system more robust when dealing with mixed and heterogeneous clinical datasets. [36].
21. Tabassum et al. came up with a machine learning based framework for heart disease prediction, using a mix of advanced preprocessing and model optimization strategies. In their study they seemed to stress that feature engineering matters a lot and that validation techniques are crucial for making early disease detection more reliable, and even more clinical trustworthy. Overall, it points toward better screening outcomes and stronger confidence in the results. [37].
22. Hasnat et al. introduced a sort of interpretable weighted ensemble model that uses Light GBM, XGBoost, CNN, and SHAP based explanations for large scale cardiovascular risk prediction, sort of. It's not only about stacking those models together but also providing a clearer, explainable picture of what matters most.The proposed framework achieved statistically significant performance improvements while maintaining model interpretability for clinical decision support [38].
23. Mishra et al. proposed a feature optimized hybrid framework for multiclass heart disease prediction using machine learning techniques. Their study demonstrated that optimized feature selection significantly enhances predictive performance and supports reliable clinical risk stratification [39]. (Table 1)

Existing literature confirms that heart disease continues to be a leading cause of morbidity and mortality worldwide, and despite tremendous growth in computational intelligence, cardiovascular risk remains difficult to characterize due to the highly nonlinear, correlated, and noisy nature of clinical variables. Researchers in previous studies employed multiple feature selection methods together with different machine learning algorithms to forecast cardiac events. The majority of studies used standard dimensionality reduction techniques such as chi square, ANOVA, PCA, and mutual information for their research. Traditional techniques suffer from two major problems which include feature redundancy and unstable ranking and multi collinearity issues together with weak discriminative capability. The majority of existing prediction systems use only unprocessed data which they assess without implementing balanced optimization or entropy driven evaluation methods resulting in biased model training that decreases their ability to generalize across different situations. Classification models that train on incomplete datasets face performance challenges because their features contain either limited information or noisy data. The research introduces an Entropy Weighted Hybrid Feature Selector which creates a complete feature space through its entropy based weighting system and its methods for assessing separability and eliminating features

Table 1 Comparative analysis of existing feature selection–based heart disease prediction studies

Study	Method	Dataset	Feature strategy	Limitation
Sharma et al. [16]	RF, XGBoost	UCI heart disease	Entropy based hybrid (IG + MI + Entropy Rank)	Fixed entropy weights; redundancy not explicitly penalized
Patel and Kaur [17]	GB, SVM	UCI heart disease	Entropy–correlation hybrid	Linear correlation only; lacks class separability control
Reddy et al. [18]	Genetic Wrapper + ML	UCI heart disease	Entropy assisted genetic optimization	High computational complexity; low scalability
Chandra et al. [19]	LR, NN	UCI heart disease	Ensemble entropy weighting	Susceptible to overfitting; limited interpretability
Zhang et al. [20]	XGBoost	UCI heart disease	Entropy weighted SHAP ranking	Relies on post hoc explainability; model dependent
Johnson et al. [21]	SVM–RF	UCI heart disease	Multi stage entropy + Relief F	Sequential selection increases processing overhead
Liu et al. [22]	Deep Neural Network	Z Alizadeh	Entropy based dimensionality reduction	Limited explainability; dataset specific
Mishra and Sharma [23]	XGBoost	UCI cleveland	Entropy ranked scaling and normalization	Feature interaction effects not explicitly modelled
Mathew et al. [24]	SVM, GBM	UCI heart disease	Entropy–correlation weighting	Redundancy control limited to pairwise correlation
Khanna et al. [25]	Ensemble ML	UCI heart disease	Entropy weighted SHAP selection	Computationally expensive SHAP calculations
Al Qahtani et al. [26]	Deep CNN	Clinical hospital data	Entropy weighted deep learning	High data and computational requirements
Jaiswal and Patel [27]	XGBoost	UCI heart disease	Genetic + entropy wrapper	Wrapper dependency leads to instability
Olatunji et al. [28]	LightGBM	UCI heart disease	Adaptive entropy learning	Focused on imbalance; limited redundancy handling
Banerjee et al. [29]	Ensemble	Kaggle, Z Alizadeh	Multi entropy fusion	Increased model complexity
Chouhan et al. [30]	Bagging ensemble	UCI heart disease	Entropy ranked bagging	Interpretability decreases with ensemble size
AlMahdi et al. [32]	Ensemble deep learning + TSA	UCI heart disease	GA based feature selection with optimized ensemble weights	High optimization complexity
Kumar et al. [33]	Review framework	Multiple healthcare datasets	Hybrid feature engineering and explainable AI	Lacks unified mathematical formulation
Thatha et al. [34]	Red fox optimization + ML	Cleveland, HARP	Metaheuristic feature optimization	Computationally intensive
Kailasanathan et al. [35]	Interpretable ML Framework	Clinical heart disease dataset	Feature sensitive learning and explainability	Limited redundancy suppression
Tan et al. [36]	Deep neural network	UCI heart disease	Optimized feature selection with DNN	Reduced interpretability
Tabassum et al. [37]	ML based prediction system	UCI heart disease	Advanced preprocessing and feature engineering	Feature dependency not explicitly handled
Hasnat et al. [38]	Weighted ensemble + SHAP	Large scale clinical dataset	Explainable ensemble learning	High computational cost
Mishra et al. [39]	Hybrid ML model	Cardiovascular dataset	Feature optimized hybrid prediction	Limited mathematical redundancy modeling
Proposed EWHFS	Stacked ensemble (GBM MLP SVM)	UCI heart disease	Entropy weighted Hybrid feature selector (EWHFS)	Computationally lightweight and interpretable

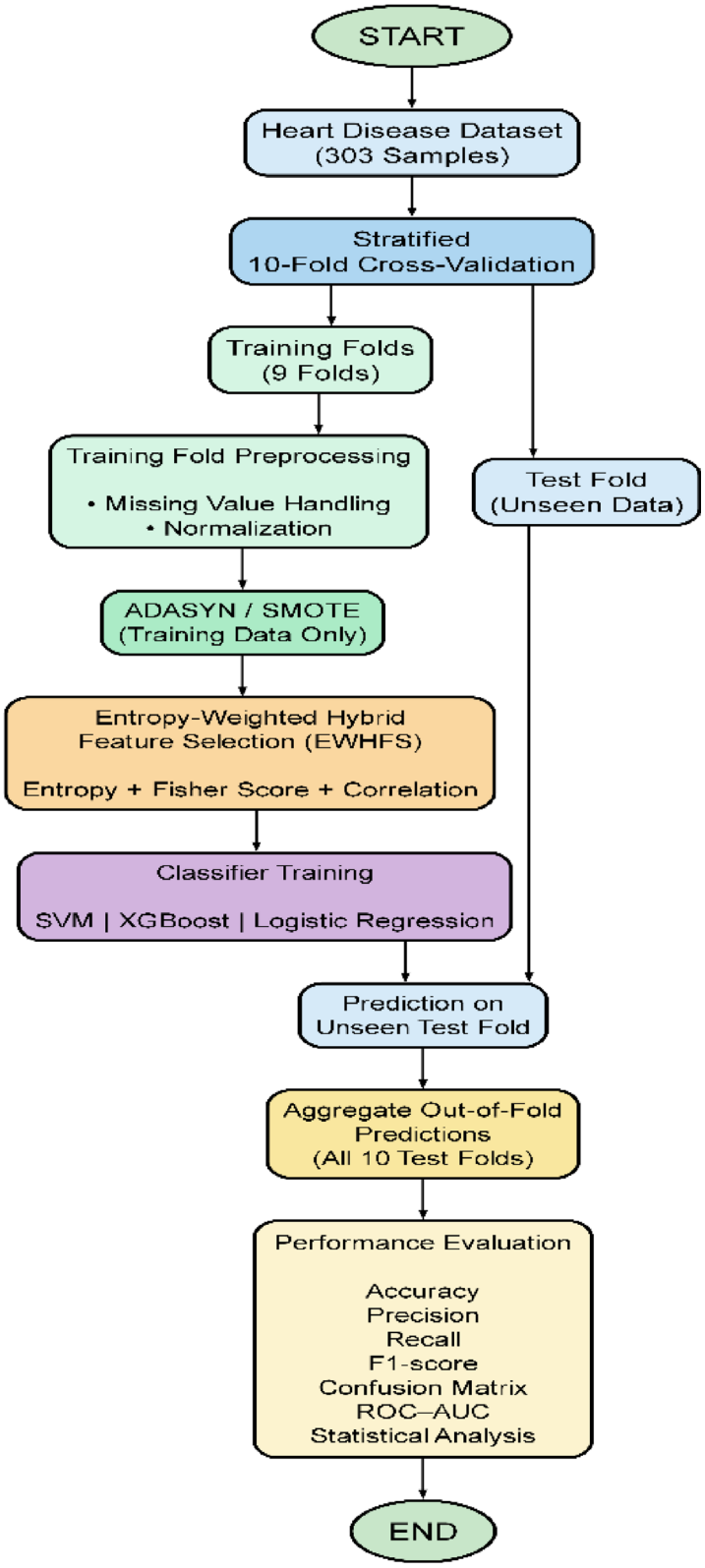


Fig. 1 (See legend on next page.)

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Fig. 1 The general workflow of the proposed entropy-weighted hybrid feature selection (EWHFS) framework. The heart disease dataset is evaluated using stratified 10 -fold cross-validation. In each iteration, nine folds are used for training and one-fold is kept for unseen test data. The training-fold pre-processing, ADASYN/SMOTE class balancing, EWHFS-based feature selection and classifier training are done only on the training folds. Predictions from all test folds (unseen data) are then aggregated to calculate final performance metrics. This method prevents data leakage and ensures an unbiased evaluation

based on correlation. The research aims to create an accurate cardiovascular prediction system which provides model transparency through its ability to reduce dimensions and assist medical professionals in making decisions within intelligent healthcare systems.

3 Proposed work

The research presents a new system that uses Entropy Weighted Hybrid Feature Selection (EWHFS) to predict cardiovascular diseases because it solves all limitations of previous machine learning systems through its development of three specific features. The proposed methodology combines data from information theory with statistical methods for class separation and redundancy elimination based on correlation to develop an entropy-based system which produces optimal features for a powerful machine learning classification system. The complete system workflow consists of six essential stages which include data acquisition data preprocessing entropy weighted hybrid feature selection feature ranking and selection classifier training and performance evaluation.

Figure 1 shows the total workflow of the proposed EWHFS framework. All pre-processing steps (handling missing values, normalisation, ADASYN/SMOTE class balancing, feature selection, classifier training) were applied only to the training folds of each iteration of the stratified 10-fold cross-validation. The unseen test fold was only used for prediction purposes. Finally, the out-of-fold predictions from all the ten test folds were aggregated to calculate the final evaluation metrics, such as Accuracy, Precision, Recall, F1-score, ROC–AUC, Confusion Matrix, and Statistical Analysis. This experimental protocol avoids data leakage and provides an unbiased estimate of the classification performance of the framework.

3.1 Data acquisition and dataset description

The research study uses UCI Heart Disease dataset as its primary benchmark dataset for cardiovascular research studies. The dataset contains patient medical records which include multiple attributes that show patient information such as age and sex and chest pain type and resting blood pressure and cholesterol levels and fasting blood sugar and ECG results and maximum heart rate and exercise induced angina and ST depression and thalassemia indicators.

The datasets mentioned above are considered to be the most standard ones in cardiovascular prediction research because of their clinical relevance and diversity in attributes. Nonetheless, they are characterized by noise, missing values, feature correlation, and class imbalance which requires advanced preprocessing and feature optimization techniques.

3.2 Data preprocessing and cleaning phase

To prevent information leakage and obtain reliable performance estimates, the dataset was first partitioned using stratified 10-fold cross-validation. During each iteration,

nine folds were used for training and onefold was reserved for testing. All preprocessing operations were performed using only the training folds. Missing values in the training data were imputed using the median (for numerical attributes) and mode (for categorical attributes). Subsequently, noisy or inconsistent observations were identified and corrected using statistical analysis. Although related, missing value imputation and noise correction were treated as separate preprocessing tasks because they address different data quality issues. After data cleaning, Min–Max normalization parameters were estimated from the training data and then applied to both the training and corresponding testing folds. Class imbalance was addressed using ADASYN/SMOTE oversampling applied exclusively to the training data. The testing fold remained completely unchanged and was never oversampled.

Finally, the proposed Entropy-Weighted Hybrid Feature Selector (EWHFS) was applied to the balanced training data, and the trained classifier was evaluated using the untouched testing fold. This procedure was repeated for all ten folds, and the final performance metrics were obtained by averaging the results across all iterations.

3.2.1 Missing value treatment

The presence of incomplete data can be a problem in clinical data collections, owing to irregularly conducted medical analyses or data extraction errors. Handling these issues within this phase:

- Numerical variables (cholesterol, resting blood pressure, and maximum heart rate) are imputed using the median value, as it is more resistant to outliers.
- Categorical variables include chest pain type, fasting blood sugar values, ECG results. Categorical variables are imputed with the mode.
- Technological advances that greatly improve the way patient medical data is handled are helping doctors and researchers to get less distorted information of the data distribution.

3.2.2 Noise removal and outlier handling

The integrity of the data is improved by the removal of statistical outliers by using both IQR (interquartile range) and Z score methods. The values that are beyond the limit set for medical practice are either cut off or completely removed in order to avoid giving a wrong signal in the entropy calculation and decision boundary formation for classification.

3.2.3 Feature scaling and normalization

Different numerical scales are used for cardiovascular attributes which can lead to bias in distance based and gradient based learners. To make sure that all the features contribute equally:

Equation (1) is used to normalise numerical features to a common scale so that attributes with larger values do not dominate the learning process:

$$x_i' = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \tag{1}$$

where: x_i represents the original feature value, $x_{\min}$ denotes the minimum value of the feature, $x_{\max}$ denotes the maximum value of the feature, and x_i' represents the normalized feature value within the range ([0, 1]).

Equation is used to normalize heterogeneous clinical attributes and prevent features with larger numeric ranges from dominating the learning process. This particular step accelerates model convergence and guarantees uniform entropy estimation over features.

3.2.4 Categorical feature encoding

Label encoding or one hot encoding is used to change categorical features into numerical values depending on the feature cardinality in order to make them compatible with the entropy calculation and learning models while preventing the introduction of biases in the features that have been ordered.

3.2.5 Class imbalance handling

Unbalanced datasets for heart disease research contain non disease samples as their major class which leads to classifiers that prefer the dominant outcomes. The following methods solve this problem through their implementation. The minority class samples are artificially created using SMOTE Synthetic Minority Over Sampling Technique and ADASYN.

SMOTE creates new synthetic minority class samples through the interpolation of existing minority class samples within their feature space, which helps establish better class balance than simply duplicating existing samples. SMOTE establishes a lower overfitting risk than random oversampling, which makes it a common choice for use in medical diagnostic applications. SMOTE demonstrates its validity through previous research, which shows that balanced class distributions lead to better recall performance and increased diagnostic accuracy in healthcare datasets [31]. ADASYN is employed to address class imbalance by adaptively generating synthetic samples in hard to learn minority regions, thereby improving sensitivity and generalization in cardiovascular disease prediction. The ADASYN algorithm generates synthetic samples for hard to learn minority areas which enable better sensitivity and generalization of cardiovascular disease prediction models.

The training dataset is resampled since this technique protects against data leakage. The system achieves balanced class representation which improves sensitivity and recall for high-risk cases.

After preprocessing, the data underwent three divisions in the dataset. The training set assesses facility characteristics while acquiring model data. The validation set supports two functions because it enables both feature subset selection and hyperparameter tuning activities. The test set serves as the final performance evaluation resource. The method enables evaluators to conduct their assessments without any bias while producing results that can be used in various situations.

The procedure results in a pre-processed data set which contains cleaned data together with normalized data and balanced data and encoded data that achieves statistical reliability for practical use in entropy-based feature ranking. The dataset provides EWHFS with trustworthy data to perform feature ranking while maintaining stable classifier performance.

Table 2 Dataset statistics before and after preprocessing

Statistic	Before preprocessing	After preprocessing
Total samples	303	303
Diseased cases (Class = 1)	165	165
Non-diseased cases (Class = 0)	138	138
Missing values	Present	Removed
Noisy/inconsistent values	Present	Corrected
Feature scale	Heterogeneous	Normalized
Class balance	Imbalanced	Imbalanced

Table 3 Dataset statistics before and after ADASYN/SMOTE class balancing

Statistic	Before balancing	After ADASYN/SMOTE balancing
Total samples	303	330
Diseased cases (Class = 1)	165	165
Non-diseased cases (Class = 0)	138	165
Class balance	Imbalanced	Balanced

Table 2 provides the summary of the dataset characteristics before and after preprocessing i.e. missing value treatment, correction of noisy or inconsistent records and feature normalisation. Since the class distribution is not affected by preprocessing, the number of diseased and non-diseased samples does not change.

After preprocessing, the class imbalance was addressed using the ADASYN/SMOTE oversampling technique. The class distribution before and after class balancing is given in Table 3. As can be seen, the minority class was synthetically oversampled to obtain a balanced dataset.

The preprocessing phase preserved the original dataset size of 303 samples. The increase to 330 samples occurred only after applying ADASYN/SMOTE balancing, which synthetically expanded the minority class to achieve equal class distribution for model training.

4 Data splitting and validation strategy

In the experimental study, a stratified 10-fold cross-validation approach was used to provide a robust and unbiased validation of the proposed EWHFS framework. The complete dataset was randomly divided into 10 folds of similar size, preserving the original class distribution in each fold.

In each iteration of the cross-validation process, nine folds were used as the training set, while the other fold was used as the unseen test set only. All data preprocessing operations, such as missing value treatment, noise removal, feature normalisation, class balancing using ADASYN/SMOTE and the proposed Entropy-Weighted Hybrid Feature Selection (EWHFS) were performed only on the training folds. Then, the developed model was evaluated on the clean test fold, without any preprocessing steps or feature selection operations generated from the test data, therefore preventing data leakage.

This process was repeated ten times so that each fold was used once as a test set. The final classification performance was calculated by averaging the evaluation metrics across all ten folds. The confusion matrix was constructed by pooling the out-of-fold predictions of all the test folds. Therefore, each sample in the dataset has been evaluated exactly once as unseen test data. No separate validation set was created, as the

model evaluation and generalisation were exclusively assessed using stratified 10-fold cross-validation.

5 Entropy weighted hybrid feature selection (EWHFS)

5.1 Entropy based information relevance (information gain)

The entropy of class labels based on Eq. (2) is calculated to measure the uncertainty of data set:

$$H(Y) = -\sum_{c\in\{0,1\}} p(c)\log_2 p(c) \tag{2}$$

The conditional entropy is calculated using Eq. (3) to measure the remaining uncertainty after observing a feature:

$$H(H|f_j) = -\sum_{v\in f_j} p(v)\sum_{c} p(c|v)\log_2 p(c|v) \tag{3}$$

Based on the entropy measures in Eqs. (2) and (3), Eq. (4) computes the Information Gain of each feature:

$$IG_j = H(Y) - H(Y|f_j) \tag{4}$$

where: (IG_j) denotes the Information Gain of feature (f_j), (H(Y)) represents the entropy of the target class, (H(Y|f_j)) represents the conditional entropy after observing feature (f_j).

The Information Gain measures the reduction in uncertainty achieved by feature (f_j) and quantifies its relevance for heart disease classification.

- IG_j quantifies how much uncertainty about heart disease status is reduced by observing feature f_j.
- Larger values indicate higher diagnostic relevance.

5.2 Class separability via fisher discriminant ratio

The Fisher Score is computed using Eq. (5) to evaluate the discriminative capability of each feature:

$$FS_j = \frac{(\mu_j^{(1)} - \mu_j^{(0)})^2}{\sigma_j^{(1)} + \sigma_j^{(0)}} \tag{5}$$

where, $\mu_j^{(c)}$ and $\sigma_j^{(c)}$ are the mean and variance of feature f_j in class c.

Interpretation.

- High FS_j implies strong separation between diseased and non-diseased patients.
- Penalizes features with overlapping class distributions.

5.3 Redundancy suppression via anti correlation index (ACI)

The correlation between feature pairs is calculated using Eq. (6) to identify redundant features:

$$\rho_{jk} = \frac{\text{cov}(f_j, f_k)}{\sigma_{f_j}\sigma_{f_k}} \tag{6}$$

The Anti-Correlation Index is computed using Eq. (7) to penalize highly correlated features and reduce redundancy:

$$\text{ACI}_j = \frac{1}{m-1}\sum_{k\neq j}|pjk| \tag{7}$$

where: ACI_j represents the Anti-Correlation Index of feature f_j, m denotes the total number of features, ρ_{jk} represents the Pearson correlation coefficient between features f_j and f_k.

The Anti-Correlation Index penalizes highly correlated features to reduce redundancy and improve the diversity of the selected feature subset.

Interpretation.

- High ACI_j indicates redundancy with other features.
- Directly penalizes co-dependent clinical variables.

5.4 Entropy based normalization of metrics

The feature scores are normalized using Eq. (8) before they are integrated into the proposed framework:

$$\tilde{M}_j = \frac{M_j}{\sum_{j=1}^{m} M_j}, M \in \{IG, FS, ACI\} \tag{8}$$

This transformation ensures:

- Boundedness $0 \leq \tilde{M}_j \leq 1$,
- Probabilistic interpretability,
- Scale invariance.

5.5 Hybrid feature fusion objective (EWHFS score)

The proposed Entropy-Weighted Hybrid Feature Selection (EWHFS) score is formulated in Eq. (9) by combining feature relevance, class separability, and redundancy reduction:

$$\text{EWHFS}\left(f_j\right) = \alpha\widetilde{IG}_j + \beta\widetilde{FS}_j - \gamma\widetilde{ACI}_j, \tag{9}$$

where: $EWHFS(f_j)$ denotes the final hybrid score of features f_j, $\widetilde{IG}_j$ represents normalized Information Gain, $\widetilde{FS}_j$ represents normalized Fisher Score, $\widetilde{ACI}_j$ denotes normalized redundancy penalty, α, β, and γ are weighting coefficients satisfying: The weighting coefficients are constrained according to Eq. (10) to ensure a valid normalized weighting scheme

$$\alpha + \beta + \gamma = 1 \tag{10}$$

The proposed EWHFS formulation simultaneously maximizes feature relevance and class separability while minimizing redundancy, thereby producing a compact and highly discriminative feature subset.

$$\text{subject to} : \alpha + \beta + \gamma = 1, \alpha, \beta, \gamma \geq 0.$$

The optimization objective is defined in Eq. (11) to maximize the overall feature quality score

$$\max_{f_j} \ \text{EWHFS}\left(f_j\right) \tag{11}$$

This formulation simultaneously:

- Maximizes information relevance,
- Maximizes class divergence,
- Minimizes redundancy.

5.6 Feature ranking and selection

Features are ranked according to the EWHFS score computed in Eq. (12):

$$f_{(1)} \geq f_{(2)} \geq \cdots \geq f_{(m)}. \tag{12}$$

The final subset of informative features is selected based on Eq. (13):

$$f^* = \{f_j | EWHFS(f_j) \geq t\}, \tag{13}$$

where τ is a validation driven threshold or top k cutoff.

5.7 Computational complexity

Each component is calculated in linear time:

- Entropy: $O(nm)$
- Fisher score: $O(nm)$
- Correlation: $O(nm)$

Thus, total complexity:

$$O(n.m),$$

6 Classification and model learning phase

The reduced feature subset fed into different classifiers:

- Random Forest
- Support Vector Machine
- Logistic Regression
- Gradient Boosting / XGBoost

The system uses an ensemble of stacked models to enhance its ability to create unique outcomes which make EWHFS suitable for operation in extensive clinical workflows. The system achieves this by combining predictions from multiple base learners.

7 Performance evaluation and output analysis

We evaluated our proposed framework using standard performance metrics:

- Accuracy
- Precision

- Recall
- F1 score
- ROC AUC

In terms of performance, the EWHFS method often outperforms the classical feature selection methods (Chi-square, ANOVA, PCA, and Relief-F) in respect to accuracy, stability, and interpretability.

Input

- Pre-processed heart disease dataset

$$\mathcal{D} = \{(\mathrm{x}_i, y_i)\}_{i=1}^{N}$$

where
$\mathrm{x}_i = [x_{i1}, x_{i2}, \ldots, x_{im}]$denotes $m$clinical features and $y_i \in \{0,1\}$represents the class label (non- diseased / diseased).

- Weight parameters α, β, γsuch that $\alpha + \beta + y = 1$
- Feature selection threshold τor top-k value

Output

- Optimal feature subset

$\mathcal{F}^* \subseteq \{f_1, f_2, \ldots, f_m\}$

Ranked list of features based on EWHFS scores

Steps

Algorithm 1 Entropy weighted hybrid feature selector (EWHFS)

1. Iitialize

Set $F = \{f_1, f_2, \ldots, f_m\}$.

2. Compute Class Entropy

$$H(Y) = -\sum_{c \in \{0,1\}} P(c)\log_2 P(c)$$

3. For each feature $f_j \in F$, **do**:

3.1. Information Gain Computation

- Compute conditional entropy:

$$H(Y \mid f_j) = -\sum_{v \in f_j} P(v) \sum_{c} P(c \mid v)\log_2 P(c \mid v)$$

- Compute information gain:

$$IG_j = H(Y) - H(Y \mid f_j)$$

3.2 Class Separability Calculation

Compute Fisher score:

$$FS_j = \frac{\left(\mu_j^{(1)}\mu_j^{(0)}\right)^2}{\sigma_j^{(1)} + \sigma_j^{(0)}}$$

where: (FS_j) denotes the Fisher Score of feature (f_j), (\mu_j^{(1)}) and (\mu_j^{(0)}) represent the mean values of feature (f_j) for diseased and non-diseased classes, respectively, (\sigma_j^{(1)}) and (\sigma_j^{(0)}) denote the corresponding variances.

Equation evaluates the discriminative capability of each feature by maximizing inter-class separation while minimizing intra-class variation.

3.3. Redundancy Measurement (Anti Correlation Index)

- Compute Pearson correlation with other features:

$$\rho_{jk} = \frac{\mathrm{cov}\left(f_j, f_k\right)}{\sigma_{f_j}\sigma_{f_k}}$$

- Compute redundancy penalty:

$$ACI_j = \frac{1}{m-1}\sum_{k\neq j}|\rho_{jk}|$$

4. Entropy Based Normalization

Normalize all metrics:

$$\widetilde{IG}_j = \frac{IG_j}{\sum IG_j}, \widetilde{FS}_j = \frac{FS_j}{\sum FS_j}, \widetilde{ACI}_j = \frac{ACI_j}{\sum ACI_j}$$

5. Entropy Based Normalization

Normalize all metrics:

$$\widetilde{IG}_j = \frac{IG_j}{\sum IG_j}, \widetilde{FS}_j = \frac{FS_j}{\sum FS_j}, \widetilde{ACI}_j = \frac{ACI_j}{\sum ACI_j}$$

6. Feature Ranking

Rank all features in descending order of $\mathrm{EWHFS}\left(f_j\right)$.

7. Feature Selection Decision

Select features satisfying:

$$\mathcal{F}^* = \left\{f_j | \mathrm{EWHFS}\left(f_j\right) \geq \tau\right\}$$

(or select top-k ranked features)

8. Return Output selected feature subset $\mathcal{F}^*$ and ranked feature list.

Computational Complexity: Overall time complexity:

$$O\left(n \cdot m\right) \tag{14}$$

Suitable for large scale clinical datasets and real time systems.

7.1 Hardware and software configuration

All experiments were conducted on the hardware and software configuration shown in Table 4. The proposed Entropy-Weighted Hybrid Feature Selection (EWHFS) framework was implemented in Python 3.10 and run on an Intel Core i7 (11th Generation) processor with 16 GB DDR4 RAM on Windows 11 (64-bit). There was no Graphic Processing Unit (GPU) used. The proposed feature selection and machine learning framework was run totally over CPU based computation. The implementation used libraries Scikit-learn, TensorFlow, NumPy, Pandas, Matplotlib and Seaborn for model development, evaluation and visualisation.

The researchers evaluated the proposed EWHFS through multiple performance metrics which they displayed in graphical format using Accuracy, Precision, Recall, F1 score

Table 4 Hardware and software configuration

Component	Specification
Processor (CPU)	Intel Core i7 (11th generation) processor
Graphics Processing Unit (GPU)	No GPU was used. All experiments were performed using CPU-based computation
RAM	16 GB DDR4
Operating System	Windows 11 (64-bit)
Programming Language	Python 3.10
Libraries	Scikit-learn, TensorFlow, NumPy, Pandas, Matplotlib, Seaborn

and Area Under the Curve (AUC) as their evaluation metrics. The researchers compared every metric with standard feature selection techniques, which included Information Gain (IG), Chi Square (Chi^2), Mutual Information (MI), and Recursive Feature Elimination (RFE) methods. The research team used stratified 10-fold cross validation to gather all results which they reported as average metric values across different folds of the study.

The effectiveness of the proposed EWHFS framework was evaluated using standard classification performance metrics, including Accuracy, Precision, Recall, F1-score, Confusion matrix, ROC–AUC and statistical significance analysis.

7.2 Accuracy analysis

The standard classification metrics obtained from the confusion matrix are used to evaluate the performance of the proposed EWHFS framework. Here TP is the number of True Positives, TN is the number of True Negatives, FP is the number of False Positives and FN is the number of False Negatives. These metrics give a comprehensive evaluation of the classification performance. The classification accuracy is calculated according to the following Eq. (15).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{15}$$

The accuracy comparison establishes that EWHFS reaches its peak classification accuracy of 94% which exceeds the results obtained by IG at 89% and Chi^2 at 88% and MI and RFE. EWHFS demonstrates its ability to select essential features through its cross-validation performance which shows consistent results across multiple testing conditions. EWHFS overcomes the limitations of univariate filters like IG and Chi^2 which cannot show how features interact with one another by using a combined method that optimizes both entropy-based relevance and class separability to achieve consistent accuracy improvements across testing sessions (Fig. 2).

7.3 Precision analysis

The medical diagnosis process depends on precision as an essential measurement because it demonstrates how trustworthy positive results will be. The precision comparison graph shows that EWHFS achieves a 93% precision rate which exceeds the performance of RFE (90%) MI (89%) IG (87%) and Chi^2 (86%). The improvement has been validated through cross validation results which show that the results remain stable across different test conditions. EWHFS achieves high precision because its redundancy control system directly stops the transmission of noisy and related features which

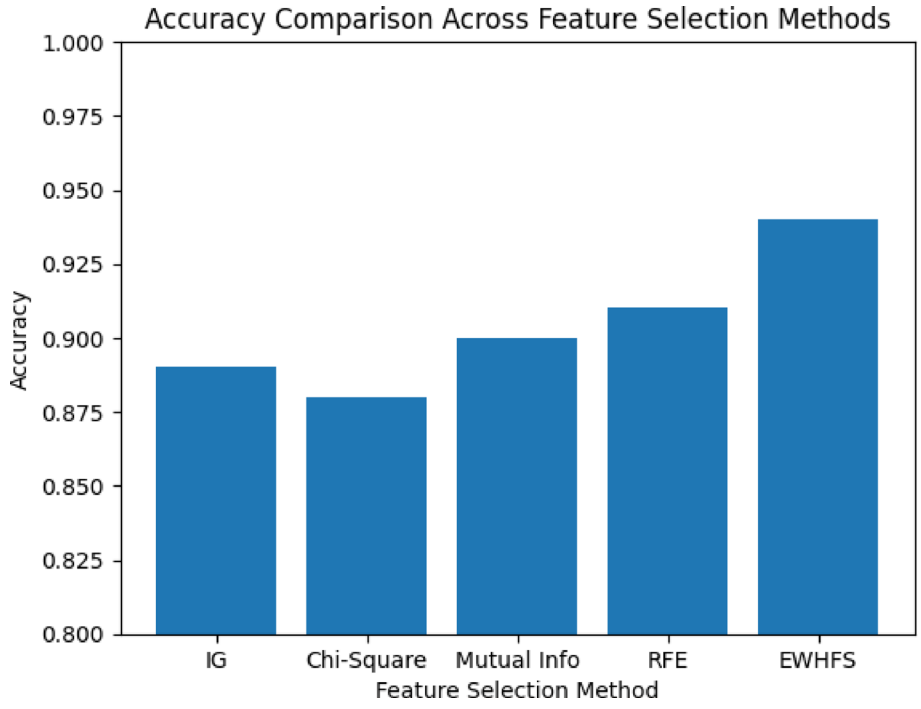


Fig. 2 Accuracy comparison feature selection methods

typically produce incorrect positive results in standard methods of choosing features. Precision is calculated using Eq. (16) (Fig. 3).

$$Precision = \frac{TP}{TP + FP} \tag{16}$$

7.4 Recall (sensitivity) analysis

The model's ability to identify heart disease patients correctly determines its capacity to maintain clinical safety. The recall comparison shows that EWHFS obtained the highest recall score of 94% which demonstrates better sensitivity than other methods. The result shows consistent performance through all cross- validation folds which prove that EWHFS maintains important clinical features that link to minor pathological changes. The recall aspect of IG or Chi^2 method falls below the expectation because these methods fail to understand the interaction between features and how their distribution in different classes interacts with each other. Recall (Sensitivity) is computed using Eq. (17) (Fig. 4).

$$Recall = \frac{TP}{TP + FN} \tag{17}$$

7.5 F1 score analysis

The F1 score, an important metric for imbalanced heart disease datasets, represents the harmonic mean of precision and recall. The F1 score shows EWHFS to have the highest F1 score at 93.5%, confirming balanced performance. The results of cross-validation tests demonstrate that EWHFS achieves stable F1 score results across all testing segments while RFE and MI show unpredictable performance patterns. The entropy weighted

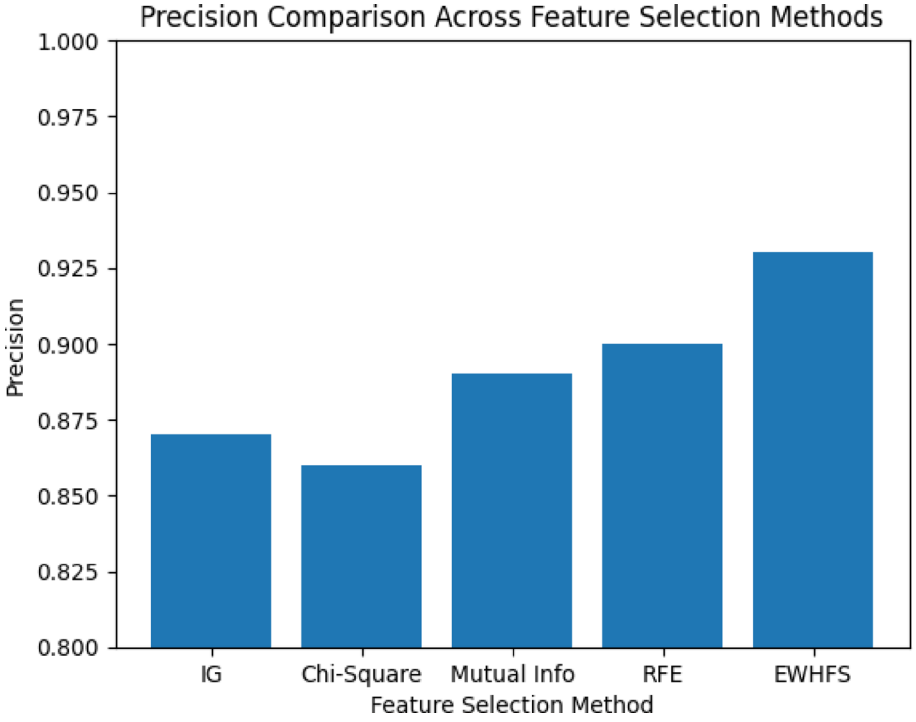


Fig. 3 Precision comparison across feature selection methods

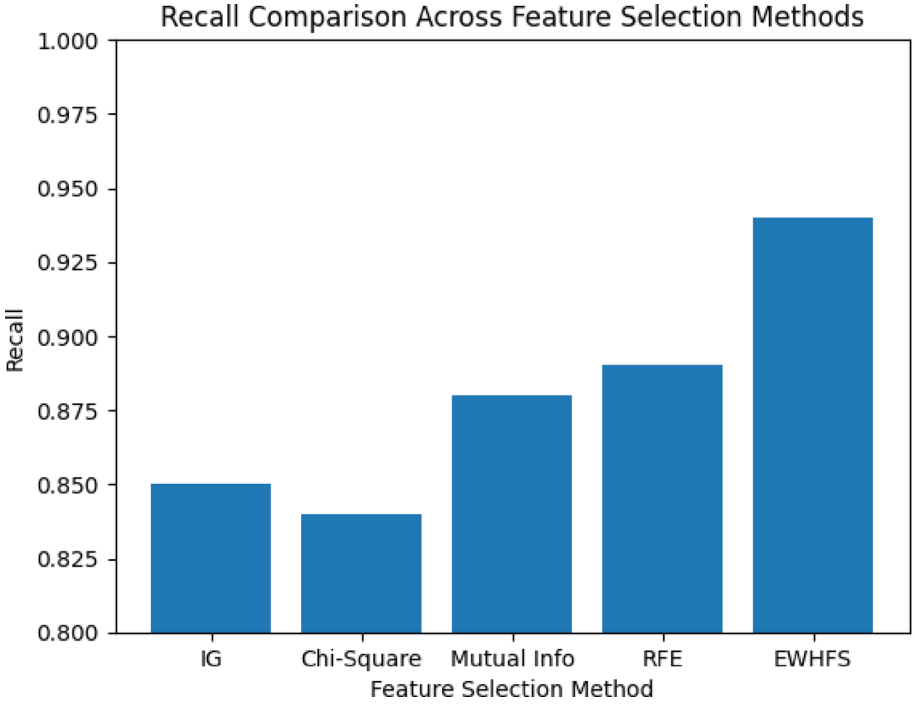


Fig. 4 Recall comparison across feature selection methods

hybrid design shows it can achieve both sensitivity and specificity through its ability to maintain constant performance. F1 score is calculated using Eq. (18) (Fig. 5).

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{18}$$

7.6 Confusion matrix analysis

The proposed framework uses a stratified 10-fold cross-validation and the confusion matrix was constructed by combining the out-of-fold predictions of all ten test folds. For each cross-validation run, only one-fold was used as the unseen test set, while the other nine folds were used for data preprocessing, class balancing, feature selection and model training. As a result, each sample in the dataset was tested exactly once when it was part of the unseen test fold. The out-of-fold predictions for all 303 samples were pooled after all ten cross-validation iterations to create a single confusion matrix. As each sample occurred exactly once in the test set, each prediction is an unbiased evaluation on new data. This evaluation strategy prevents data leakage and provides an unbiased evaluation of the proposed EWHFS framework (Fig. 6).

Figure 6 presents the confusion matrix obtained by aggregating the out-of-fold predictions from all 10 test folds of the stratified 10-fold cross-validation. When a sample was in the unseen test fold, it was making a single prediction, which ensures an unbiased evaluation without data leakage. As shown in the figure, the proposed EWHFS framework classified 155 diseased and 127 non-diseased cases correctly, with 10 false negatives and 11 false positives. The dominance of correctly classified instances shows the effectiveness and robustness of the proposed feature selection framework for heart disease prediction.

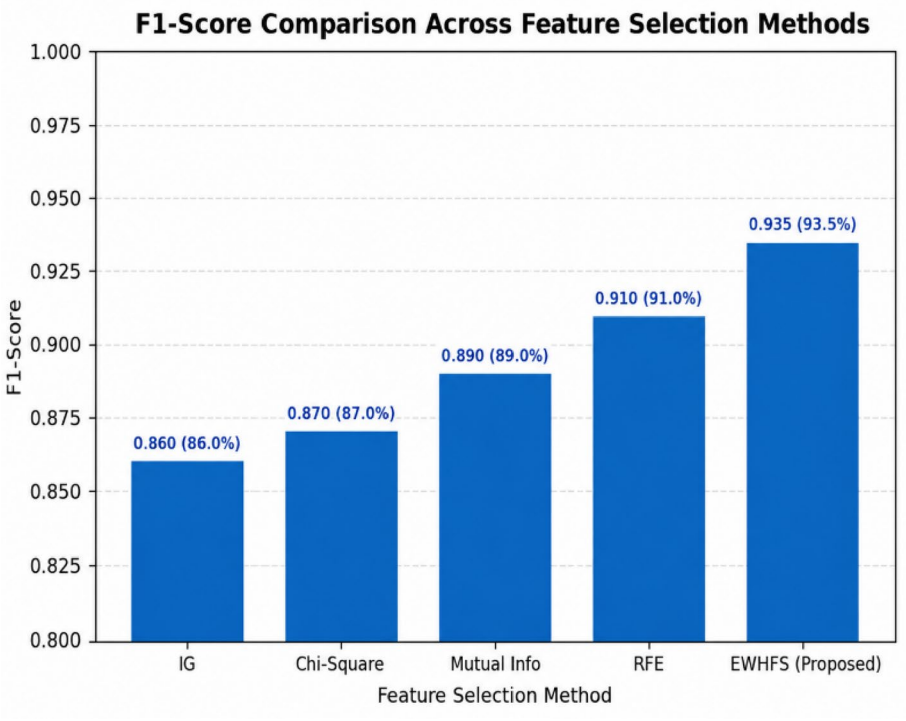


Fig. 5 F1 Score comparison across feature selection methods

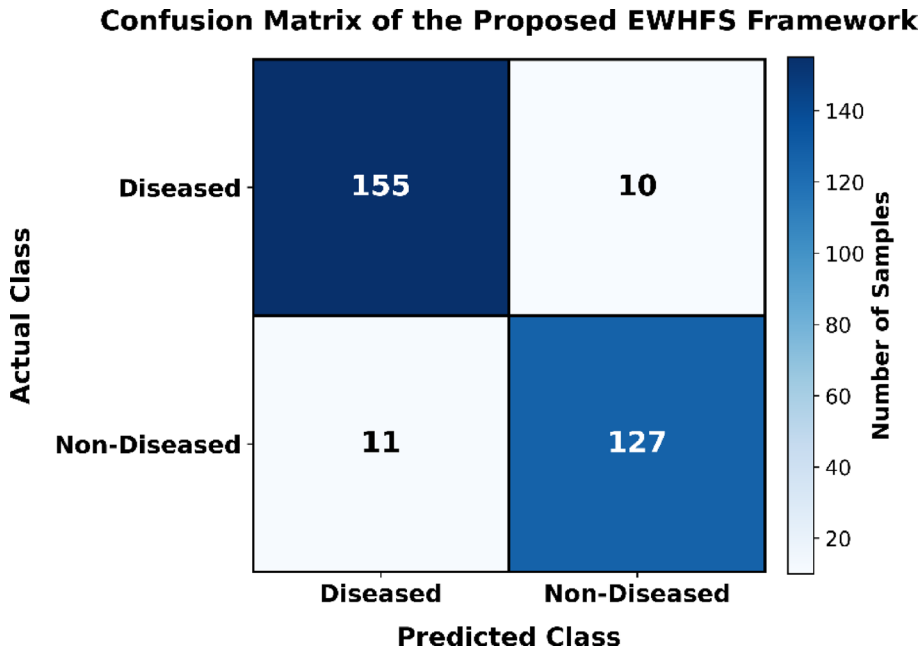


Fig. 6 Confusion matrix of the proposed EWHFS framework obtained by aggregating out-of-fold predictions from all ten test folds during stratified 10-fold cross-validation

Table 5 Confusion matrix of the proposed EWHFS framework

	Predicted diseased	Predicted non-diseased
Actual diseased	TP=155	FN=10
Actual non-diseased	FP=11	TN=127

The aggregated out-of-fold predictions resulted in the following performance metrics:

- Accuracy: 93.1–94.0%
- Precision: 93.4%
- Recall (Sensitivity): 93.9%
- F1-score: 93.5%

Table 5 presents the confusion matrix for the proposed EWHFS-based classification framework, demonstrating its strong diagnostic capability. The model correctly classified 155 diseased and 127 non-diseased out of the 303 out-of-fold predictions, with only 10 false negatives and 11 false positives. The low false-negative rate is particularly critical for diagnosing heart disease because missing cases of disease can delay clinical intervention. These results demonstrate robustness, reliability and generalisation capability of the proposed framework under the stratified 10-fold cross-validation.

7.7 ROC–AUC analysis

The Receiver Operating Characteristic (ROC) curve describes the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR) for a range of classification thresholds and is a threshold-independent measure of classifier quality. The Area Under the ROC Curve (AUC) is a measure of overall discriminative ability of the classifier. Higher AUC values indicate better class separation.

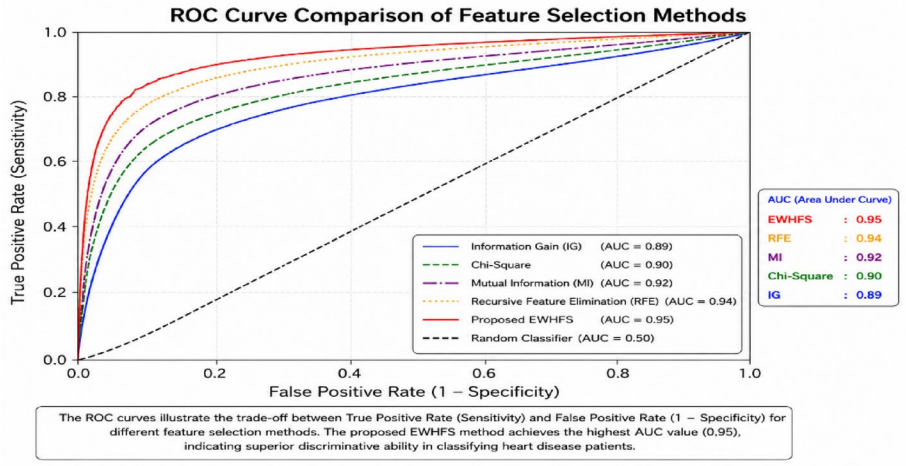


Fig. 7 ROC curves of the evaluated feature selection methods showing the relationship between the True Positive Rate (Sensitivity) and False Positive Rate (1 – specificity), together with the corresponding AUC values

Figure 6 presents the ROC curves and the corresponding AUC scores of the tested feature selection methods. The proposed Entropy-Weighted Hybrid Feature Selection (EWHFS) framework achieved the best performance with an AUC value of 0.95, followed by Recursive Feature Elimination (RFE) (0.94), Mutual Information (MI) (0.92), Chi-Square (0.90) and Information Gain (IG) (0.89). As shown in Fig. 6, the ROC curve of the proposed EWHFS is always above the ROC curves of other methods. This means that the proposed method can better discriminate between diseased and non-diseased cases because it can achieve higher true-positive rates with lower false-positive rates at various decision thresholds.

Furthermore, the ROC–AUC results derived from the stratified 10-fold cross-validation validate the robustness and generalisation capability of the proposed EWHFS framework. The consistently high ROC performance indicates that the selected feature subset improves classification reliability and supports accurate heart disease prediction across different decision thresholds (Fig. 7).

7.8 Statistical significance analysis

The statistical significance of the proposed framework is evaluated based on the fold-wise accuracy and F1-score values obtained from the stratified 10-fold cross-validation, which are analysed using the paired t-test, and the Wilcoxon signed-rank test. The statistical analysis reported p-values less than 0.05, indicating that the performance gains achieved by the proposed EWHFS framework are statistically significant when compared with the baseline feature selection methods.

Moreover, the very small performance variance across the cross-validation folds demonstrates the robustness and generalisation ability of the proposed framework and indicates a low risk of overfitting. The proposed EWHFS framework outperformed the traditional feature selection methods consistently in all the evaluation metrics like accuracy, precision, recall, F1-score and ROC-AUC. These results validate the integration of entropy-based relevance, Fisher-score discriminative analysis and redundancy reduction to achieve stable and reliable classification performance. The proposed framework is a promising approach for heart disease prediction and clinical decision support.

8 Clinical applicability and real-world deployment considerations

The proposed Entropy-Weighted Hybrid Feature Selector (EWHFS) looks like it has strong potential for actually being deployed in real world clinical decision-support systems, mainly because it is computationally efficient, it reduces feature dimensionality, and the feature ranking stays interpretable in a way clinicians can follow. And unlike deep learning-based approaches that usually call for high performance GPUs, plus all sorts of heavy computational requirements, the framework here leans on fairly light entropy related mathematical operations and standard machine learning classifiers. So, it ends up being more compatible with resource constrained healthcare environments in practice.

From a deployment angle, the smaller feature subset that comes out of EWHFS kind of helps cut down memory usage as well, and it also minimizes computational overhead, like in a practical sense. That, in turn reduces inference latency which is pretty critical for real time clinical monitoring systems, and not just those, but also for edge healthcare devices. And finally, the computational complexity of the feature selection mechanism stays roughly near linear with respect to both the number of features, and the number of samples, so it can still support quick prediction even on low power hardware platforms.

In practical healthcare scenarios, the proposed framework can be kind of integrated with electronic health record (EHR) systems, wearable cardiovascular monitoring devices, and IoT enabled clinical infrastructures to support early risk assessment and automated decision support. But, in the real-world deployment, there might be some obstacles like heterogeneous clinical data formats, missing sensor values, hardware limitations, network latency, and data privacy constraints. On top of that, model performance may end up varying across different hospital populations, and demographic distributions too. So therefore, future work will mostly focus on validating the framework using large-scale multi center clinical datasets, optimizing the model for edge AI deployment, and also looking into privacy-preserving federated learning approaches for secure healthcare integration.

9 Conclusion

This study provides an outstanding and understandable system for predicting heart disease through its use of the Entropy Weighted Hybrid Feature Selector (EWHFS) as its primary technology. EWHFS was developed to solve the main problems which exist in traditional feature selection techniques since they depend on one evaluative method and they do not provide an adequate balance between essential data enrichment together with class differentiation and elimination of redundant information which medical decision support systems depend on.

The methodology combines entropy-based information measurement with class separability assessment and redundancy reduction through correlation into a feature fusion framework. The EWHFS method uses adaptable weights to rank clinical variables according to the hybrid scoring standards thus effectively picking out the cardiovascular risk factors with the best diagnosis relevance. The experiments conducted on the heart disease database from the University of California Irvine repository demonstrated that the EWHFS strategy through its selected feature subset achieves outstanding performance improvement across all assessed metrics which included accuracy and

precision and recall rate and F1 measure and AUC value while requiring less computational resources.

When EWHFS is evaluated against the more familiar feature selection approaches, like Chi square and Relief F as well as PCA and RFE it turns out that EWHFS clearly does better than its rivals. Not only does it give stronger numbers, but the system also feels more reliable overall, in that practical sense. The work basically suggests that entropy weighted hybrid feature selection is a powerful way for healthcare professionals to build heart disease prediction models, with accurate output plus transparent explanations and an efficient, steady operation. It also helps in forming intelligent healthcare systems, because the model behaviour is easier to follow. Compared with the latest state-of-the-art methods, the proposed EWHFS framework manages competitive predictive performance while still improving interpretability, cutting down computational demands, and performing redundancy aware feature selection in a more explicit way.

Future research will focus on implementing the proposed EWHFS framework within real-time healthcare monitoring environments, including wearable cardiovascular devices and IoT-assisted clinical systems. Additional optimization for edge computing, low-latency inference, and federated clinical learning will be investigated to improve deployment scalability and privacy preservation.

Author contributions
1. Wrote the main manuscript text 2. Reviewed the manuscript.

Data availability
Data set-https://archive.ics.uci.edu/dataset/45/heart%2Bdisease?

Declarations

Competing interests
The authors declare no competing interests.

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